

Secure Communication Protocol for Unmanned Aerial Vehicles (UAVs)

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Engineering

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Abstract

The increasing deployment of Unmanned Aerial Vehicles (UAVs) in both civilian and military applications necessitate robust security mechanisms for their communication links.

The widely adopted MAVLink protocol, while lightweight and efficient, lacks built-in encryption and authentication, rendering UAV systems vulnerable to eavesdropping, spoofing, message manipulation, and replay attacks.

This project proposes and implements a secure communication protocol for UAVs by integrating a lightweight security layer based on the ChaCha20-Poly1305 authenticated encryption algorithm into a MAVLink-like protocol.

The system is designed to ensure confidentiality, integrity, and authentication between a Ground Control Station (GCS) and multiple UAVs operating in an ad-hoc wireless environment.

Using the ns-3 network simulator and the OpenSSL library, the proposed solution introduces sequence number tracking for replay protection and evaluates performance using metrics such as packet loss, latency, throughput, encryption/decryption time, and energy efficiency.

Simulation results demonstrate that the security layer effectively blocks malicious traffic, including replay and jamming attacks, while maintaining low processing overhead (91.6 μ s for encryption, 16.24 μ s for decryption) and minimal energy consumption (99.55% efficiency).

The proposed approach significantly enhances MAVLink communication security without compromising real-time operational performance.

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List of Abbreviations

Abbreviations	Definition
UAV	Unmanned Aerial Vehicles
MAVLink	Micro Air Vehicle Link
UDP	User Datagram Protocol

Introduction

1.1 Introduction

Unmanned Aerial Vehicles (UAVs), commonly known as drones, have gained significant attention in recent years due to their wide range of applications in both civilian and military domains. These applications include surveillance, aerial photography, disaster management, environmental monitoring, and logistics services. UAV systems rely on wireless communication to exchange data between the UAV and the Ground Control Station (GCS).

One of the commonly used communication protocols in UAV systems is MAVLink, which enables efficient data exchange between system components.

Despite its advantages, MAVLink was originally designed with limited security mechanisms, which may expose UAV communications to various cyber threats such as eavesdropping, spoofing, and message manipulation. Therefore, enhancing the security of UAV communication has become an important research area to ensure reliable and secure operation of these systems.

1.2 Problem Statement

UAV communication systems rely heavily on wireless networks, which makes them vulnerable to several security threats. The MAVLink protocol, widely used in UAV systems, lacks strong built-in encryption and authentication mechanisms in its basic design. This limitation may allow attackers to intercept transmitted data, inject malicious messages, or gain unauthorized control over the UAV.

As a result, ensuring secure communication between the UAV and the Ground Control Station is a critical challenge that needs to be addressed.

1.3 Project Objectives

The main objective of this project is to enhance the security of UAV communication by improving the protection mechanisms used in MAVLink-based communication systems.

The specific objectives of the project include:

- Analyzing the security vulnerabilities of UAV communication protocols.
- Studying existing security mechanisms used in UAV networks.
- Proposing an approach to enhance the security of MAVLink communication.
- Evaluating the performance and effectiveness of the proposed security solution.

1.4 Significance of the Study

The importance of this project lies in improving the security of UAV communication systems, which are increasingly used in critical applications. Enhancing communication security helps protect sensitive data, prevents unauthorized access to UAV systems, and reduces the risks associated with cyberattacks. Furthermore, this study contributes to the development of more reliable and secure UAV communication frameworks that can be applied in various real-world applications.

1.5 Conclusion

This chapter introduced the background of UAV communication systems and highlighted the importance of securing communication protocols such as MAVLink. The problem statement, project objectives, and significance of the study were also presented. The following chapters will discuss the theoretical concepts, related work, and the proposed approach for improving UAV communication security.

Chapter 1: Theoretical Concepts

1.1 Introduction

Unmanned Aerial Vehicle (UAV) systems rely on wireless communication for data exchange between different components, which makes security a critical requirement. This chapter presents the fundamental theoretical concepts related to UAV systems, communication protocols, and security mechanisms used to protect UAV communications.

1.2 Unmanned Aerial Vehicle (UAV) Systems

Unmanned Aerial Vehicles (UAVs), also known as drones, are aircraft that operate without an onboard human pilot. They can be remotely controlled or operate autonomously using embedded sensors and control algorithms. A typical UAV system consists of the UAV platform, the Ground Control Station (GCS), and the communication link.

Wireless communication enables the transmission of telemetry data and control commands. However, it exposes UAV systems to security threats such as eavesdropping and spoofing attacks [5].

1.3 MAVLink Communication Protocol

MAVLink is a lightweight communication protocol widely used in UAV systems for message exchange between UAVs and ground control stations. It is optimized for low-bandwidth environments.

However, MAVLink lacks built-in encryption and authentication mechanisms, making it vulnerable to attacks such as message injection and interception [2].

1.4 Communication Security in UAV Systems

1.4.1 Confidentiality

Confidentiality ensures that transmitted data is accessible only to authorized users. This is achieved using encryption techniques [3].

1.4.2 Integrity

Integrity ensures that data is not altered during transmission. Message authentication codes and hashing techniques are used to guarantee data integrity [4].

1.4.3 Authentication

Authentication verifies the identity of communicating entities to prevent unauthorized access [3].

1.5 Encryption in UAV Communication

1.5.1 Symmetric Encryption

Symmetric encryption uses a shared key for encryption and decryption. It is efficient and suitable for UAV systems with limited resources [1].

1.5.2 Asymmetric Encryption

Asymmetric encryption uses public and private keys and provides stronger security but requires higher computational resources [3].

Lightweight encryption techniques have been proposed to secure MAVLink communication without affecting performance [1], [2].

1.6 Secure Communication Layer

The Secure Communication Layer is introduced to enhance the security of UAV communication systems, particularly when using protocols such as MAVLink that lack built-in security features. This layer operates at the application level and is responsible for protecting data before transmission.

The Secure Layer ensures the main security requirements, including confidentiality, integrity, and authentication. It works by applying cryptographic operations such as encryption, message authentication, and secure key exchange before sending the data through the communication channel.

By integrating this layer, UAV communication becomes more resilient against common cyber threats such as eavesdropping, spoofing, and message manipulation [1], [2].

1.6.1 ChaCha20 Encryption Algorithm

ChaCha20 is a lightweight symmetric encryption algorithm providing high speed and strong security for UAV communication. It is particularly effective in resource-constrained environments like UAVs [1].

1.6.2 Poly1305 Authentication Algorithm

Poly1305 is used to ensure data integrity by generating authentication tags and is commonly combined with ChaCha20 to provide authenticated encryption. This combination ensures both confidentiality and integrity of UAV communications [1]. [2].

These algorithms, when used together, strengthen the security of UAV systems, providing confidentiality, integrity, and authentication while maintaining performance suitable for UAV platforms [1], [2].

1.7 End-to-End Security

End-to-End Security ensures that data is encrypted from the source (UAV) to the destination (GCS) and cannot be read at any intermediate point [4].

1.8 Chapter Summary

This chapter presented the fundamental concepts of UAV communication systems and their associated security challenges. The structure of UAV systems, including the interaction between UAVs, Ground Control Stations (GCS), and wireless communication links, was described.

The MAVLink protocol was introduced as a lightweight communication protocol widely used in UAV applications. Its limitations in terms of lacking built-in security mechanisms were also highlighted.

Key security requirements, including confidentiality, integrity, and authentication, were discussed, along with common cryptographic approaches such as symmetric and asymmetric encryption. A secure communication layer was proposed to enhance MAVLink security using ChaCha20 for encryption and Poly1305 for authentication.

Finally, the concept of end-to-end security was presented to ensure complete data protection during transmission. These concepts provide the theoretical foundation for the system design and implementation described in the following chapters.

Chapter 2: Literature Review

2.1 Introduction

This chapter reviews previous research related to UAV communication security, focusing on lightweight encryption, authentication mechanisms, and secure UAV protocols.

Article Title	Publication Year	Journal / Source	Authors	What Was Achieved in This Paper	Reported Performance Metrics	Strengths	Relevance to our Project
Security Strategies for Unmanned Aircraft Systems Networks	2023	IEEE ICC Workshop	Marco Hernandez; Gurkan Gur; Kamesh Namuduri	Proposed a comprehensive security framework for UAS networks and analyzed threats such as spoofing, jamming, and DoS using a Zero-Trust approach.	Threat classification, vulnerability analysis of UAV communication layers	Comprehensive overview of UAV security challenges	Helps define the threat model for securing UAV-GCS communication
Security-Enhanced Lightweight Authentication on Key-Agreement Protocol for UAV Communication	2025	Applied Sciences (MDPI)	Zhoucan He et al.	Introduced a lightweight authentication and key-agreement protocol suitable for resource-constrained UAV systems.	Computation cost, authentication delay, ECC operation count	Low computational overhead and resistance to replay/impersonation attacks	Supports designing lightweight authentication for UAV communication
Enhancing MAVLink Security in UAV Systems	2025	University Thesis / Research Study	Daniel Gomez Rey; Juan Pablo	Implemented encryption for MAVLink	Latency, RAM usage, communication	Real UAV implementation and comparison of encryption methods	Directly relevant for securing MAVLink-based UAV

			Hernandez Troncoso	communication using AES-GCM, ChaCha20-Poly1305 and TLS 1.3.	overhead, battery consumption		communication
A Secure Communication Protocol for Unmanned Aerial Vehicle Networks	2022	Conference / Research Paper	Various Authors	Designed a secure protocol with encryption and mutual authentication to protect UAV network communication.	Authentication time, key generation cost, communication overhead	Improves confidentiality and resistance to network attacks	Provides protocol-level design ideas for secure UAV communication
UAV-Based Secure Data Communication: Multilevel Authentication Perspective	2024	Sensors (MDPI)	Abdullah Aljumah	Proposed a multi-level authentication model for Internet of UAVs using hashing, encryption and XOR operations.	Latency, throughput, packet loss, energy consumption	Layered authentication with improved efficiency	Useful for designing multi-layer authentication mechanisms
A Novel Cipher for Enhancing MAVLink Security	2024	Journal / Conference in UAV Cybersecurity	Various Authors	Propose a new encryption algorithm to protect MAVLink messages and prevent data	Encryption and decryption time, and the efficiency of the algorithm.	An algorithm tailored to the MAVLink protocol	Useful for developing a lightweight encryption layer for our project

				interception or modification			
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2.2Chapter Summary

Despite significant advances in UAV security, most existing approaches either introduce high computational overhead or lack real-time evaluation in lightweight UAV environments.

This motivates the proposed ChaCha20-Poly1305-based secure communication framework integrated with MAVLink-like protocol in this work.

Chapter 3: Design and Implementation of a Secure MAVLink Communication System for UAVs

3.1 Introduction

This chapter presents the proposed secure UAV communication system. The system integrates a MAVLink-like communication protocol with a lightweight security layer based on ChaCha20-Poly1305 authenticated encryption. The aim is to ensure secure, reliable, and efficient communication between the Ground Control Station (GCS) and multiple UAVs.

3.3 Network Topology Design

The network consists of one Ground Control Station (GCS) and two UAV nodes operating in an Ad-hoc wireless environment.

An IPv4 addressing scheme (192.168.1.0/24) is used to enable communication between nodes.

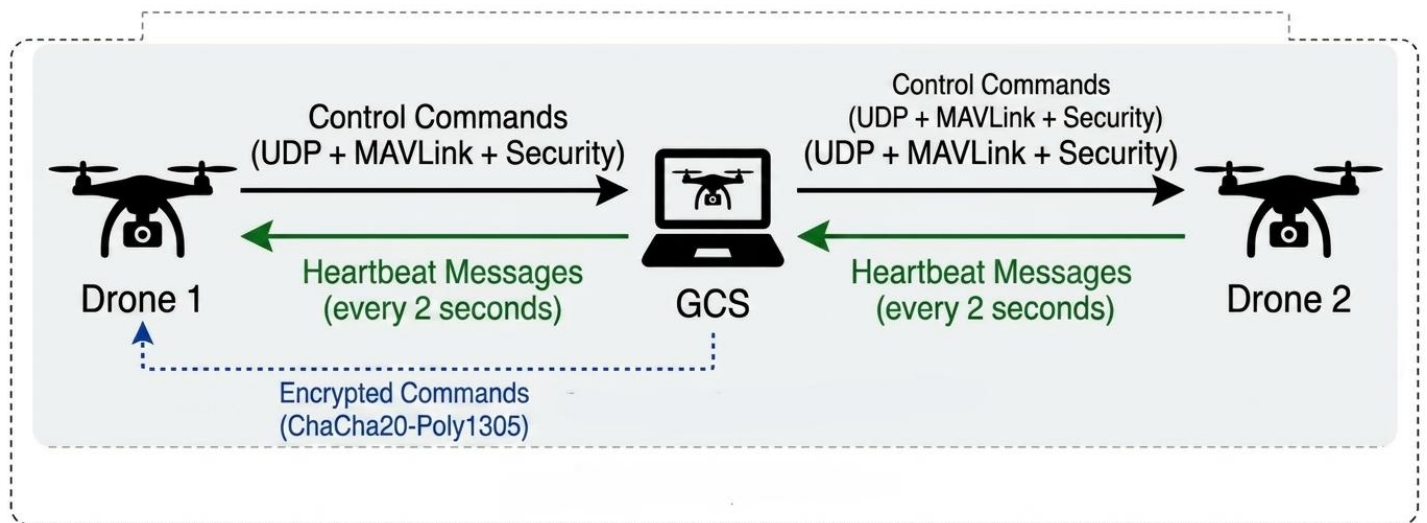


Figure 1:Network Topology Design

3.4 Implementation Parameters

The system configuration parameters are summarized as follows:

Table 1: Simulation Parameters for UAV Communication System

Parameter	Value	Description
Number of UAVs	2	Two UAV nodes
Simulation Time	12 s	Total simulation duration
Packet Size	32–64 bytes	Includes MAVLink + security overhead
Data Rate	1 Mbps	Wireless transmission rate
Transmission Interval	2 s	Packet sending interval

MAC Protocol	IEEE 802.11b	Ad-hoc wireless MAC
Transport Protocol	UDP	Connectionless transport protocol

3.5 MAVLink-like Communication Protocol

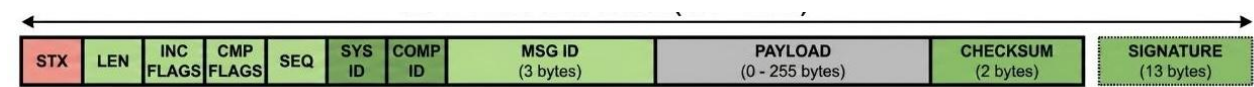


Figure 2:MAVLink Packet Structure

3.5.1 Packet Structure

Each transmitted packet follows a structured format inspired by the MAVLink protocol to ensure efficient parsing and communication reliability.

The packet structure is defined as follows:

Table 2:MAVLink Frame Structure

Field	Size	Function
STX	1 byte	Start of frame (0xFD)
LEN	1 byte	Payload length
Incompat Flags	1 byte	Non-compatible feature flags
Compat Flags	1 byte	Compatible feature flags
Seq	1 byte	Sequence number
SysID	1 byte	System ID
CompID	1 byte	Component ID
MsgID	3 bytes	Message ID
Payload	Variable	Actual data
Checksum	2 bytes	Error detection (CRC)
Signature	13 bytes (optional)	Message authentication

This structure ensures efficient transmission, error detection, and extensibility for future enhancements.

3.5.2 Message Types

The communication protocol supports a predefined set of control and status messages, which are exchanged between the GCS and UAV nodes.

These include:

- **CONN**: Used to establish a connection between UAV and GCS.
- **UP**: Command to increase altitude or move upward.
- **DOWN**: Command to decrease altitude or move downward.
- **STOP**: Command to halt UAV movement or operation.
- **HEARTBEAT**: Periodic message transmitted by UAVs to indicate operational status and connectivity.

These messages ensure continuous interaction and coordination between system entities.

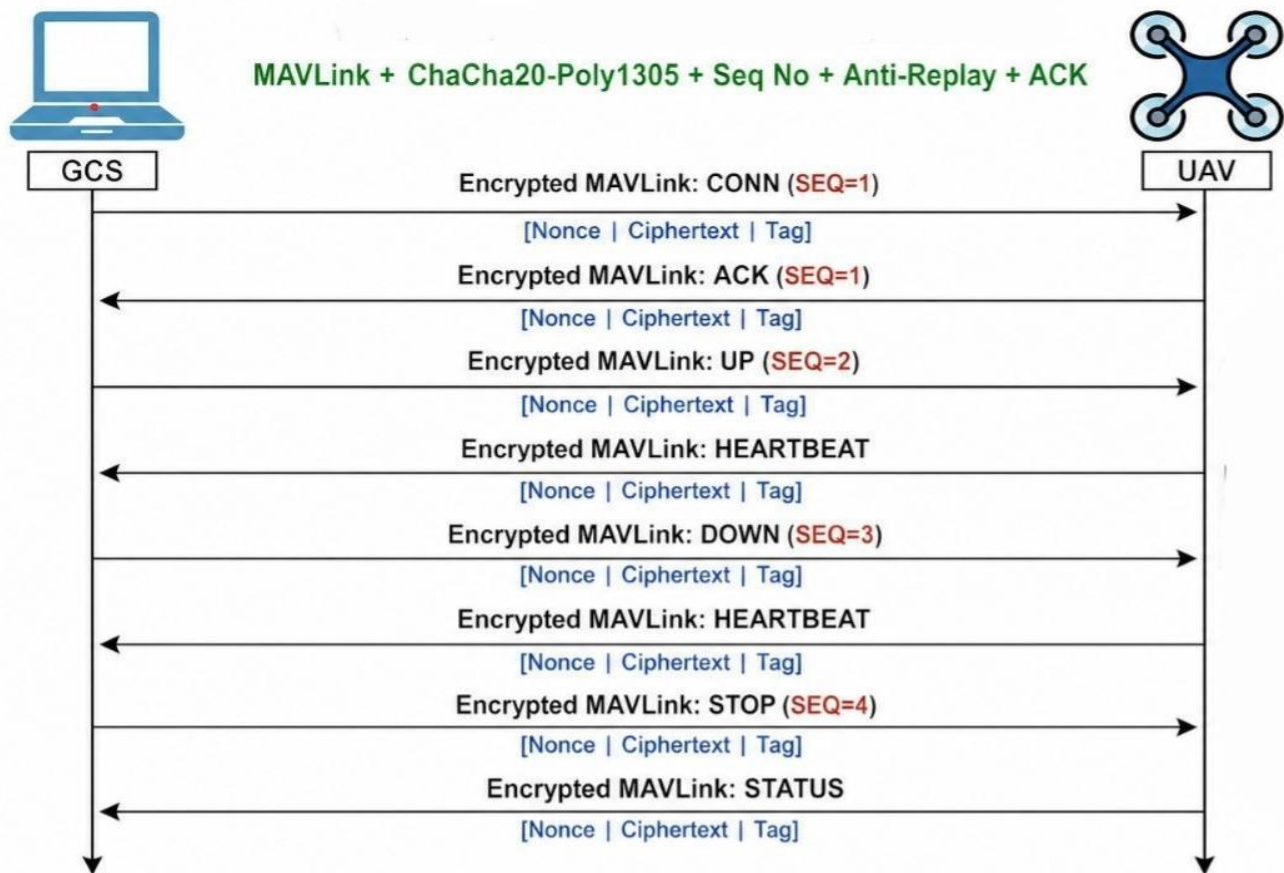


Figure3 : Secure MAVLink Communication Sequence Diagram

3.5.3 Sequence Number and Replay Protection

The Sequence Number (SEQ) field plays a critical role in ensuring communication reliability and security. It is incremented for each transmitted packet and is used at the receiver side to:

- Detect duplicate packets
- Maintain message ordering
- Prevent replay attacks
- Packets with previously received sequence numbers are discarded to ensure freshness and integrity of the communication stream.

3.6 System Architecture (OSI Model and Protocol Stack)

The system follows a layered architecture as follows:

- Application Layer: MAVLink-like Protocol
- Security Layer: ChaCha20-Poly1305 AEAD
- Transport Layer: UDP
- Network Layer: IPv4
- Data Link Layer: IEEE 802.11b Ad-hoc
- Physical Layer: Wireless Channel

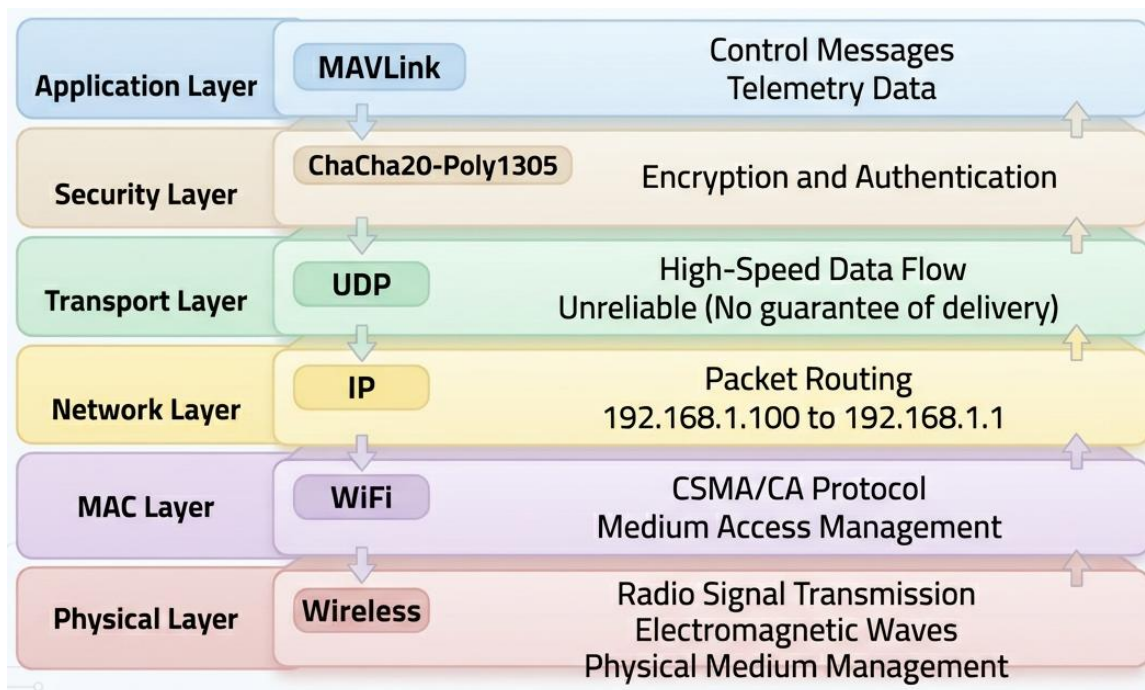


Figure 4: System Architecture (OSI Model and Protocol Stack)

3.7 Security Layer Design

The system integrates a lightweight security layer based on authenticated encryption using the ChaCha20-Poly1305 algorithm, combined with sequence number tracking for replay protection. Each transmitted packet is encrypted using ChaCha20 and appended with an authentication tag generated by Poly1305.

Upon reception, each UAV verifies the authenticity of the packet by checking the authentication tag and validates the sequence number to prevent replay attacks.

If the authentication fails or the sequence number has already been received, the packet is immediately rejected.

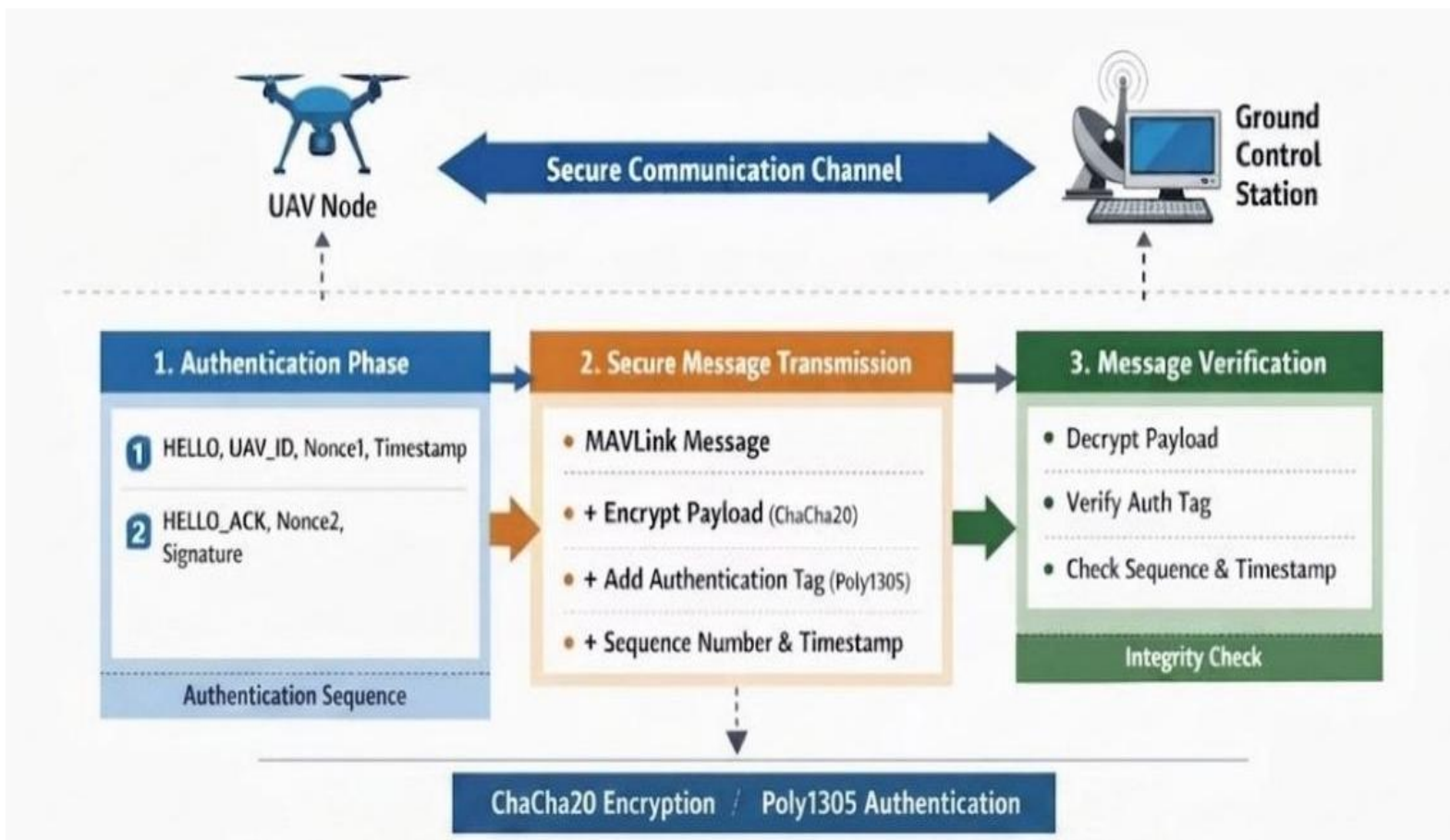


Figure 5: Secure UAV Communication Framework Based on ChaCha20-Poly1305

3.7.1 Cryptographic Model (ChaCha20-Poly1305)

Equation (1): ChaCha20 Encryption

$$\begin{aligned} (1)C &= P \oplus KS \\ KS &= ChaCha20(Key, Nonce, Counter) \end{aligned}$$

Where:

- *C*: Ciphertext
- *P*: Plaintext
- *KS*: Keystream generated by ChaCha20
- Key: secret encryption key
- Nonce: unique value used once
- Counter: block counter ensuring unique keystream blocks

Table 3:ChaCha20 Specifications

Parameter	Value	Description
Key Size	256 bits	Secret key
Nonce Size	96 bits (12 bytes)	Unique per packet
Block Size	512 bits	Internal structure
Operation	XOR	Encryption
Performance	High	Suitable for real-time

Equation (2): Poly1305 Authentication

$$(2)TAG = Poly1305(K, C, A)$$

Where:

- *TAG*: Authentication tag
- *K*: Secret key
- *C*: Ciphertext

- A: Associated data (Header, SEQ)

Table 4: Poly1305 Specifications

Parameter	Value	Description
Key Size	256 bits	Authentication key
Tag Size	128 bits (16 bytes)	Integrity verification
Output	MAC Tag	Authentication
Security	High	Prevents tampering

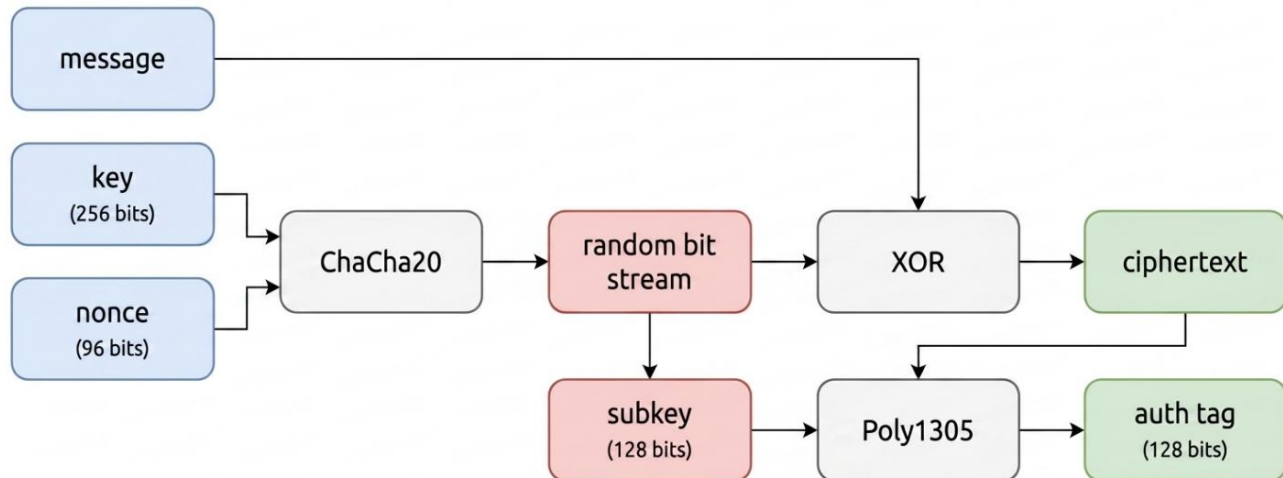


Figure 6 :ChaCha20-Poly1305 Encryption

Equation (3): AEAD Models

$$(3) (C, TAG) = AEAD_ChaCha20_Poly1305(K, Nonce, P, A)$$

The nonce (12 bytes) and authentication tag (16 bytes) are directly implemented using OpenSSL functions.

3.8 Secure Transmission and Reception Process

3.8.1 Transmission Process

The transmission process follows these steps:

- A. Create MAVLink-like message
- B. Add Sequence Number
- C. Encrypt payload using ChaCha20
- D. Generate authentication tag using Poly1305
- E. Transmit packet via UDP

3.8.2 Reception Process

The receiver performs the following steps:

Receive packet via UDP

- A. Verify authentication tag
- B. Detect replay attacks using SEQ
- C. Decrypt ciphertext using ChaCha20
- D. Parse MAVLink-like message

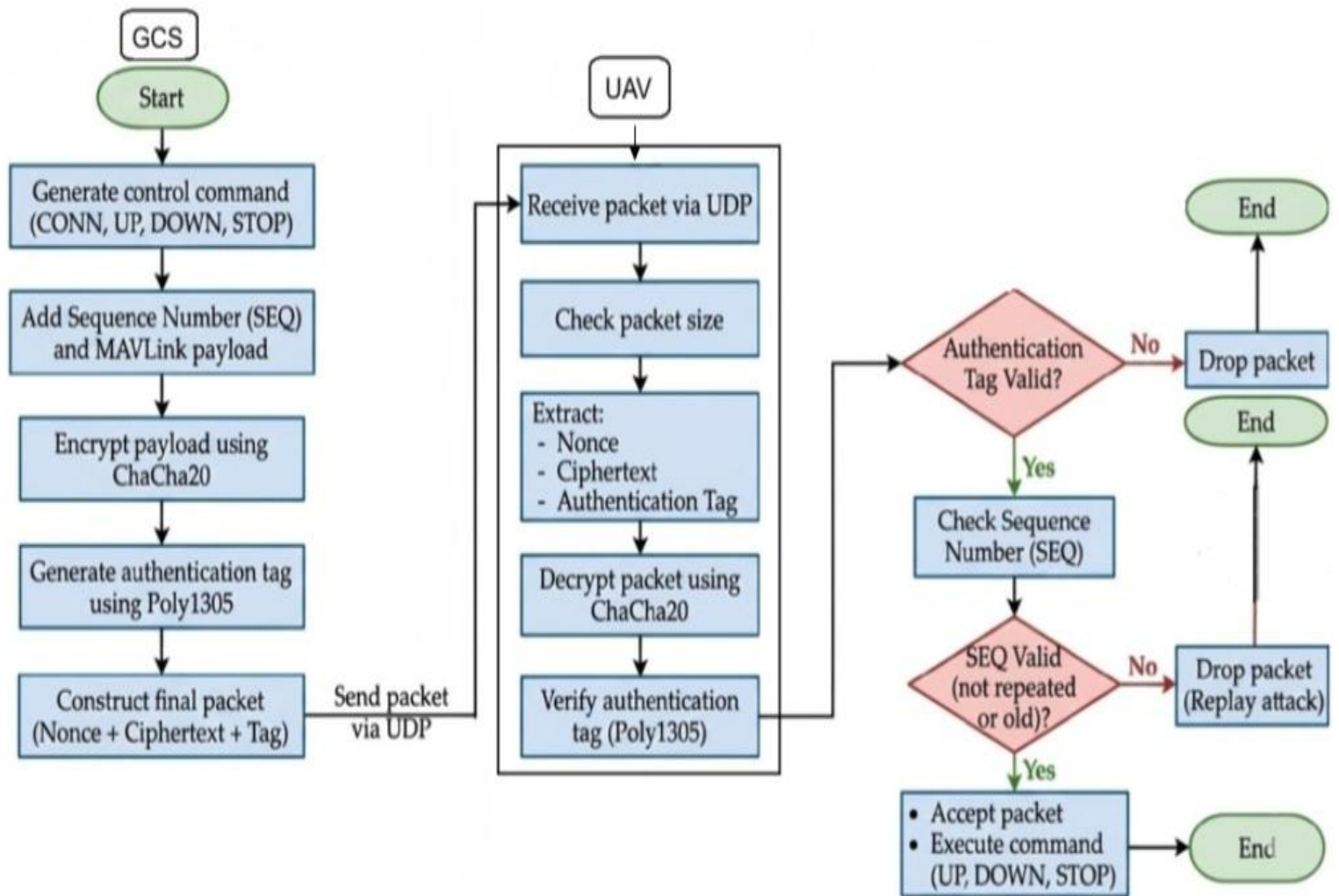
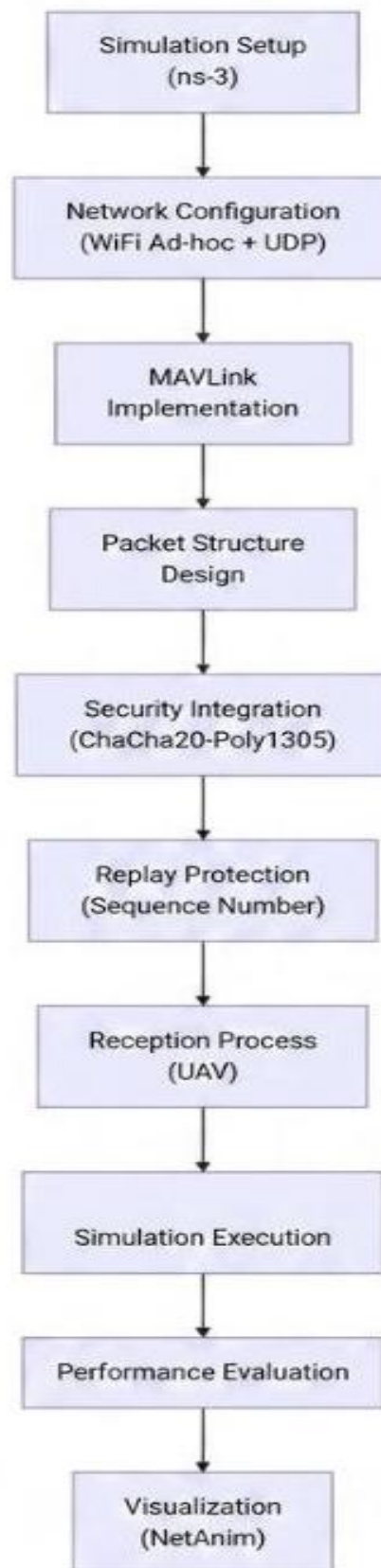


Figure 7:Secure Transmission and Reception Process

3.9 Project Implementation Phases



3.10 Communication Scenario

The system supports bidirectional communication between the GCS and UAVs.

The GCS sends control commands, while UAVs transmit telemetry data such as GPS location, battery status, and system state.

3.11 Performance Metrics

The system evaluates performance using:

Equation (4): Packet Loss Rate

$$(4)PLR = \frac{Sent - Received}{Sent} \times 100$$

Equation (5): Average Latency

$$(5)Latency = \frac{\sum Delay}{Received}$$

Equation (6): Throughput

$$(6)Throughput = \frac{Total\ Bits}{Simulation\ Time}$$

Equation (7): Replay Acceptance Rate

$$(7)Replay = \frac{Accepted\ Replay\ Packets}{Total\ Replay\ Attempts}$$

3.12 Chapter Summary

This chapter presented the design and methodology of the proposed secure UAV communication system based on a MAVLink-like protocol integrated with ChaCha20-Poly1305 authenticated encryption. The system architecture was described using a layered model following the OSI reference stack, where secure communication is ensured at the security layer through lightweight AEAD encryption combined with sequence number tracking to mitigate replay attacks.

The cryptographic foundation of the system was detailed through the ChaCha20 stream cipher for encryption and the Poly1305 algorithm for message authentication, ensuring both confidentiality and integrity of transmitted data. Furthermore, the secure transmission and reception processes were outlined, demonstrating how UAV and GCS nodes handle encryption, authentication, and replay protection during communication.

In addition, the chapter defined the overall system architecture, communication scenario, and implementation phases, providing a complete understanding of how the proposed system operates. Finally, the performance evaluation metrics, including packet loss rate, average latency, throughput, and replay acceptance rate, were introduced to assess the efficiency and reliability of the system in later experimental analysis.

Overall, this chapter establishes the theoretical and structural foundation of the system, which is implemented and evaluated in the following chapter using the ns-3 simulation environment.

Chapter 4: Practical application and Results

4.1 Introduction

This chapter presents the practical implementation of the proposed secure UAV communication system. It describes the experimental environment, software tools, simulation setup, and testing procedures used to evaluate system performance. The chapter also highlights the implementation of the MAVLink-like protocol and the integrated security mechanisms in the ns-3 simulation environment.

4.2 Implementation Environment

The system is implemented using the ns-3 network simulator, which provides a discrete-event simulation platform for wireless networks. The implementation is written in C++ and integrates cryptographic libraries for security functions.

4.3 Software Tools and Libraries

The following tools and libraries are used in the implementation:

- ns-3 Network Simulator (v3.47)
- C++ Programming Language
- OpenSSL Library (for ChaCha20-Poly1305 encryption)
- MAVLink C Library
- Net Anim (for visualization)
- Wireshark (for packet analysis)

4.4 Hardware Environment (Simulation Setup)

Although the system is simulated, the following logical hardware setup is considered:

- Ground Control Station (GCS) Node
- UAV Node 1
- UAV Node 2
- Wireless Ad-hoc communication environment (IEEE 802.11b)

4.5 Simulation Configuration

The simulation parameters are defined as follows:

Table 5: UAV Network Simulation Configuration

Parameter	Value
Number of UAVs	2
Simulation Time	12 seconds
MAC Protocol	IEEE 802.11b Ad-hoc
Mobility Model	Constant Position
Transport Protocol	UDP
Channel Type	Wireless Channel

4.6 System Implementation Overview

The system implements a MAVLink-like protocol for communication between GCS and UAVs. The main implemented functions include:

- Command transmission (CONN, UP, DOWN, STOP)
- Heartbeat messaging for UAV status monitoring
- Replay attack simulation and detection
- Packet encryption using ChaCha20-Poly1305
- Sequence number tracking for integrity protection

4.7 Communication Interface

The communication is implemented using UDP sockets in ns-3. Each UAV node has a dedicated socket to receive commands from the GCS, while periodic heartbeat messages are sent back to maintain connectivity.

4.8 Visualization and Simulation Interface

At the beginning of the simulation, UAV nodes operate in an idle state to reduce energy consumption and avoid unnecessary processing. This design choice improves overall system efficiency, especially in resource-constrained UAV environments.

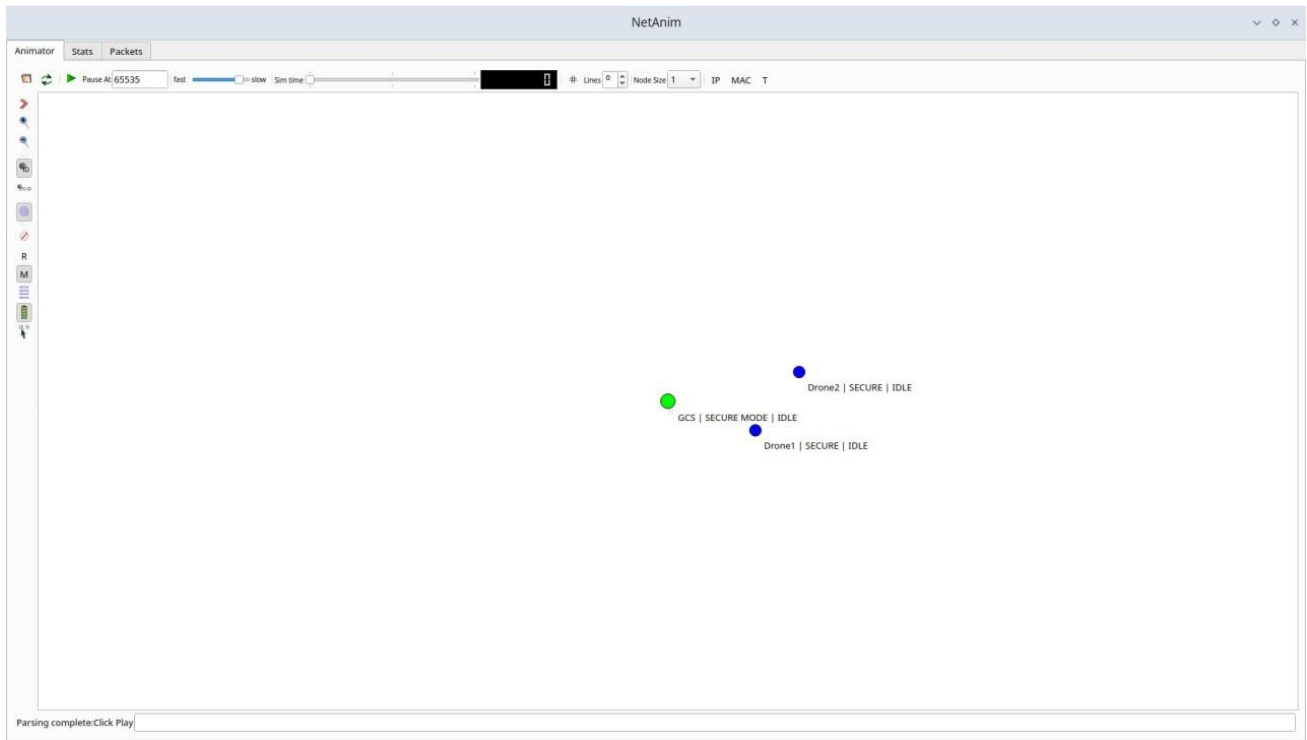


Figure 8: Initial Simulation State Showing UAV Nodes in Idle Mode

The following figure illustrates the simultaneous transmission of packets from the Ground Control Station (GCS) to multiple UAV nodes during the simulation.

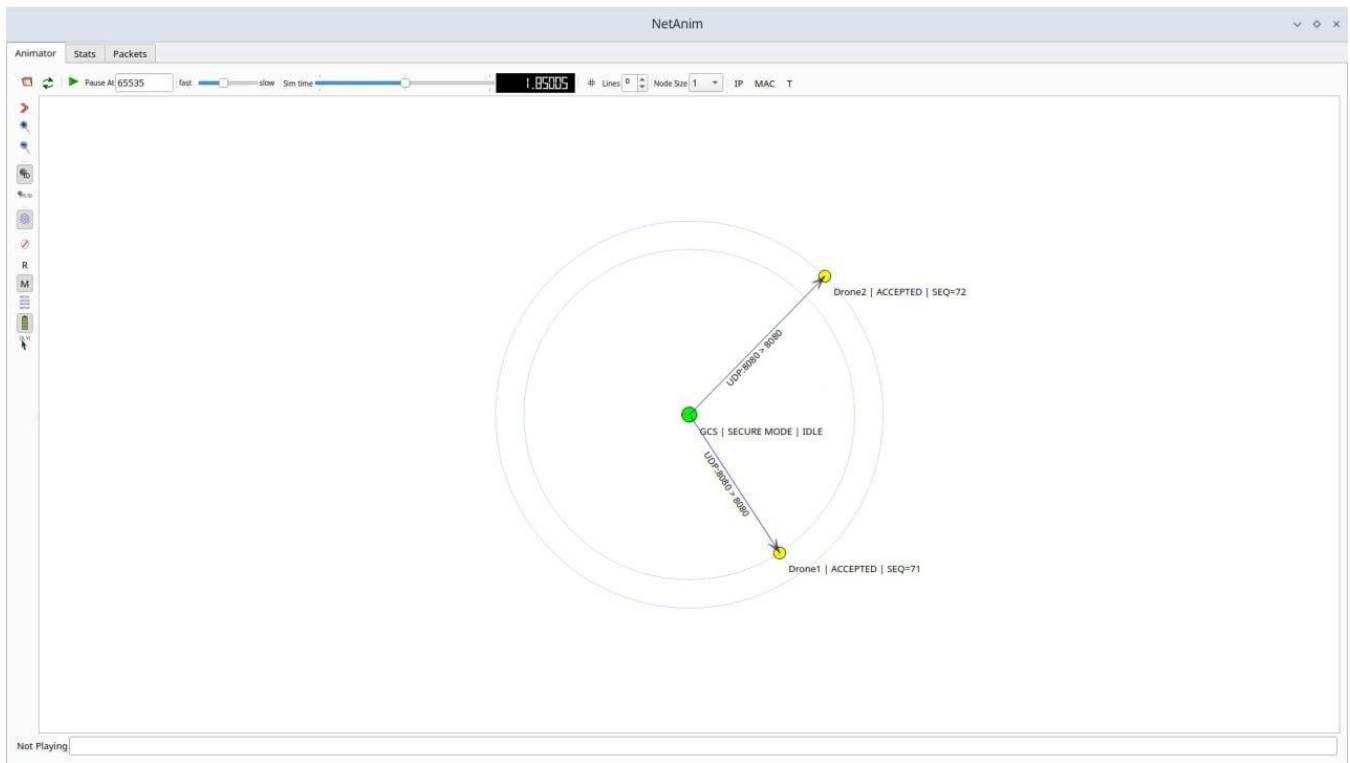


Figure 9: Simultaneous Packet Transmission from GCS to Multiple UAV Nodes

The following figure illustrates the reception of an acknowledgment (ACK) message by the Ground Control Station (GCS) from UAV 1, corresponding to a specific transmitted packet identified by its sequence number (SEQ).

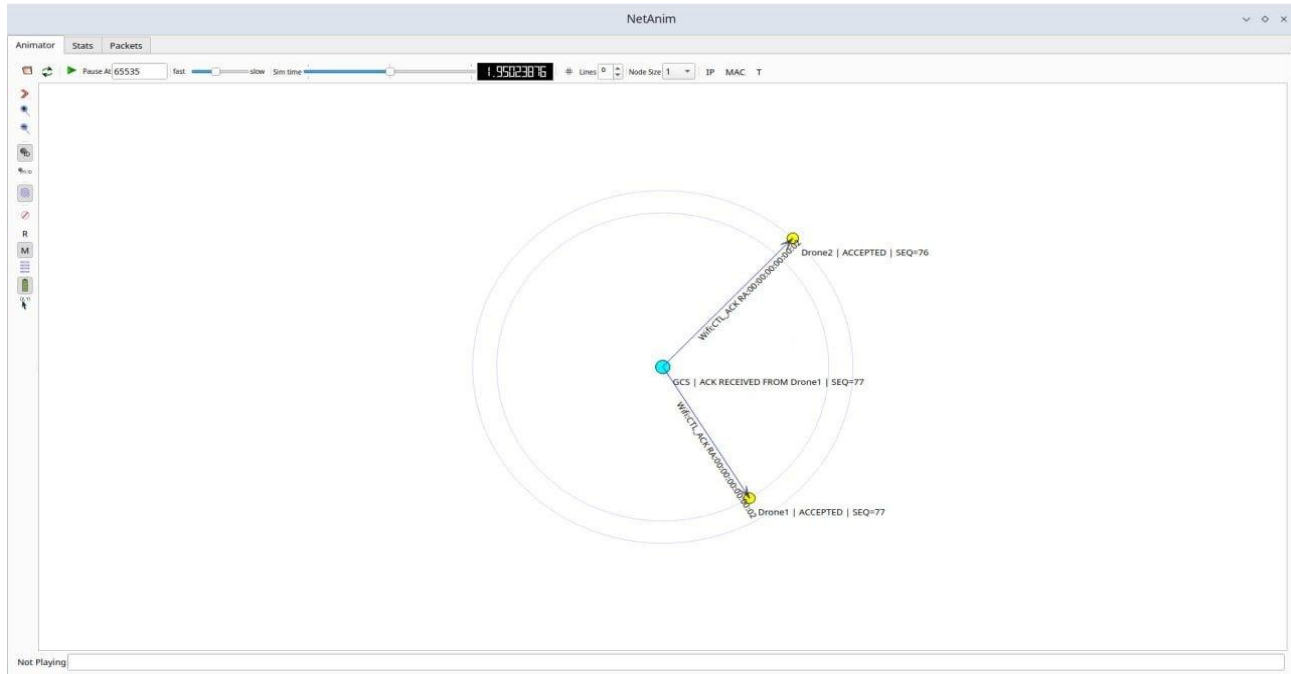


Figure 10:GCS Receiving ACK from UAV 1 with Corresponding Sequence Number (SEQ)

The following figure illustrates the addition of a fourth node representing an attacker in the network, which is used to simulate malicious activities such as jamming or packet injection.

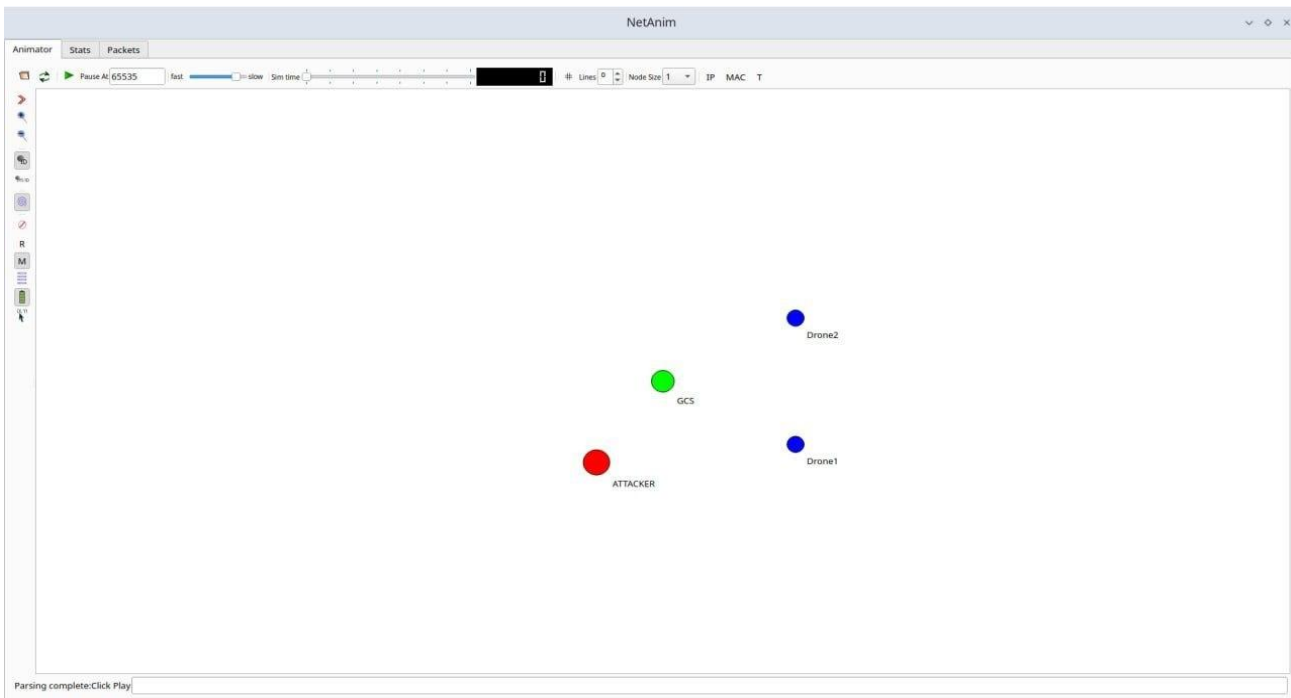


Figure 11: Network Topology with an Additional Attacker Node

The following figure illustrates the execution of a replay attack by the attacker node, where previously captured packets are retransmitted to the UAV nodes.

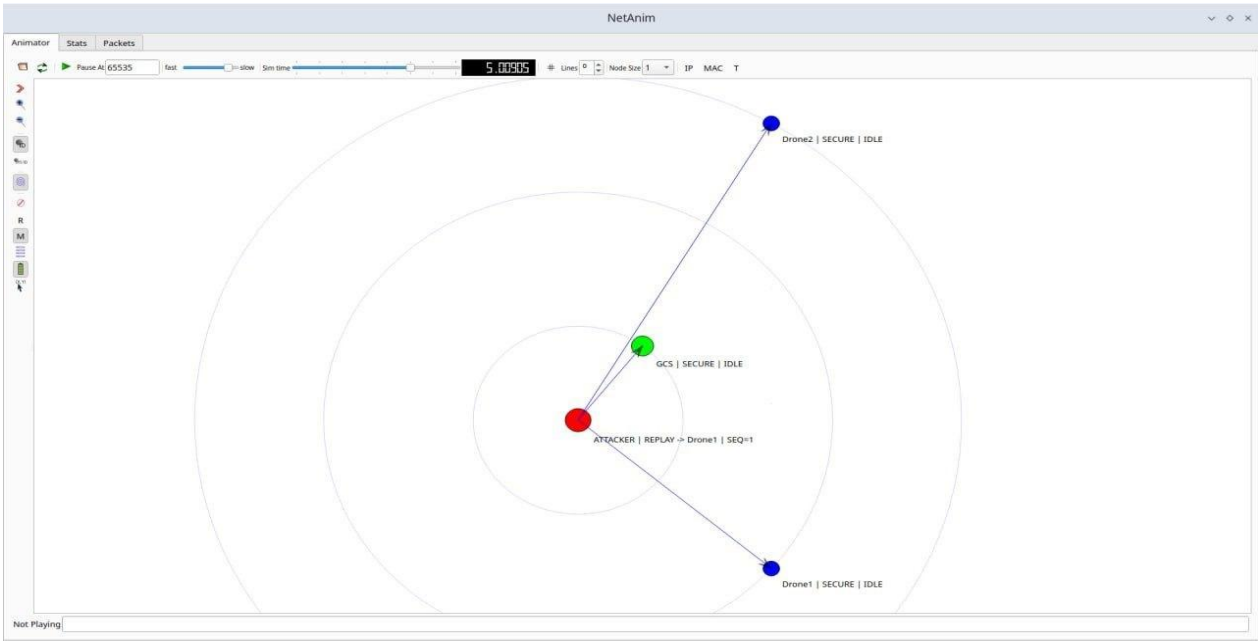


Figure12 : Replay Attack Executed by the Attacker Node on UAVs

The following figure illustrates the detection of a replay attack by UAV (1) and UAV (2), where the node identifies the malicious packet based on sequence number validation and generates a warning message.

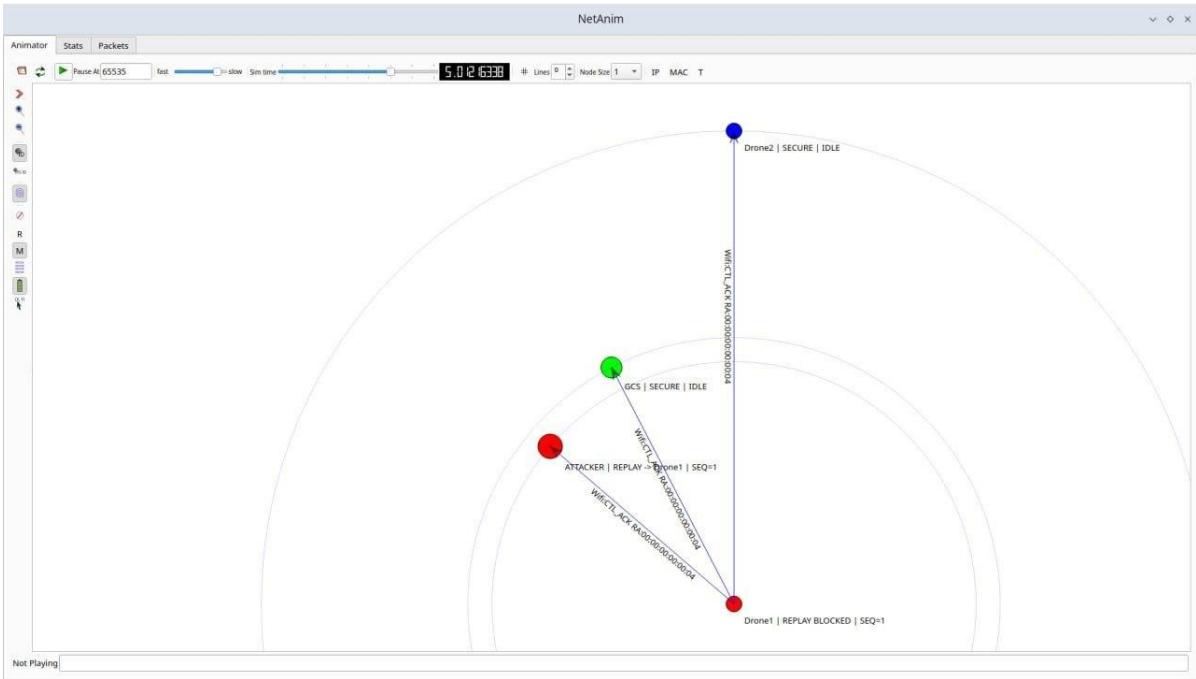


Figure13 : Replay Attack Detection by UAV 1 and Warning Message Generation

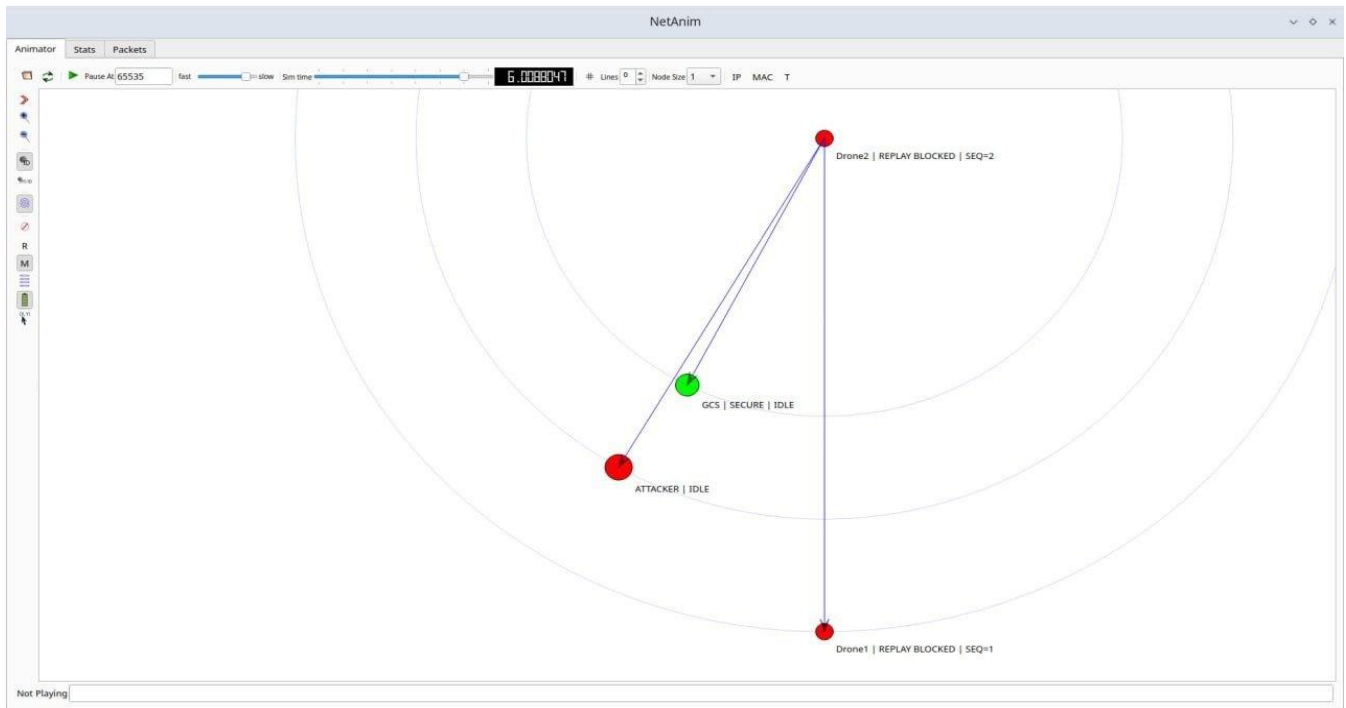


Figure14 : Replay Attack Detection by UAV 2 and Warning Message Generation

4.9 Testing Scenarios

The system is tested under different scenarios:

- Normal communication scenario (command exchange)
- Replay attack scenario
- Jamming / corrupted packet simulation
- Secure encryption/decryption validation

4.10 Performance Evaluation

The system is evaluated using the following metrics:

- Packet Loss Rate
- Average Latency
- Throughput
- Encryption/Decryption time
- Attack detection rate
- Energy efficiency

4.10.1 Results and Analysis

The following table compares the system performance before and after applying the security mechanism in terms of packet transmission and reliability.

Table 6: Performance Comparison Before and After Security Integration

Metric	BEFOR	AFTER
Original Sent Packets	100	100
Accepted Unique Packets	100	100
Packet Loss	0%	0%

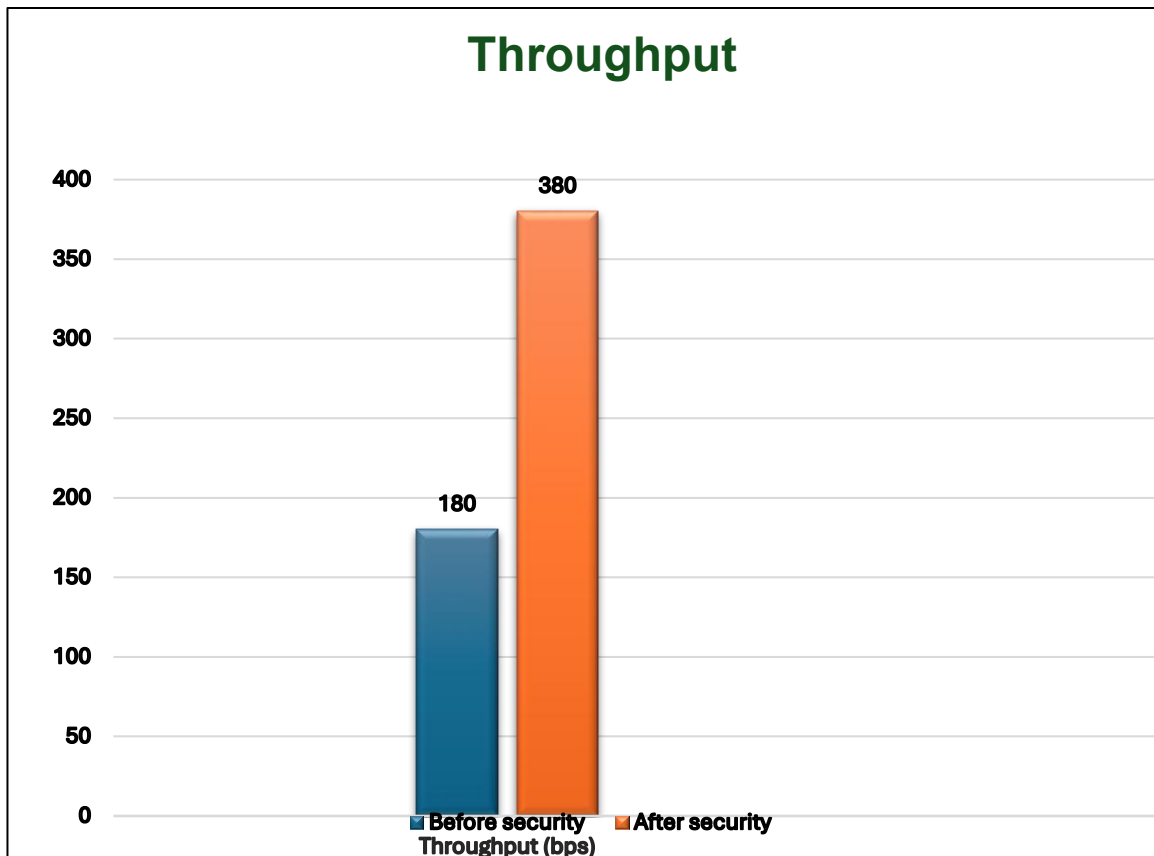


Figure15 : Throughput Comparison Before and After Security Implementation

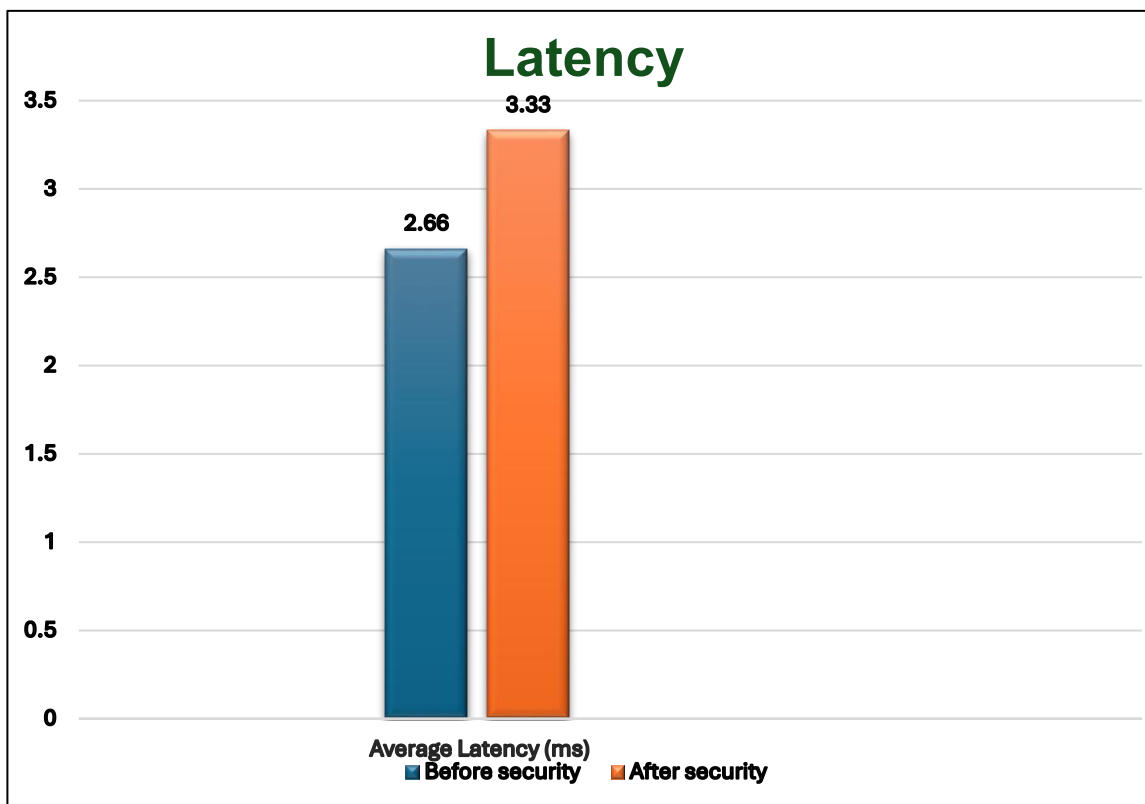


Figure16 : Throughput Comparison Before and After Security Implementation

The following table presents the security performance evaluation of the proposed system, including the number of authenticated packets and blocked malicious traffic.

Table 7: Evaluation of Security Performance Metrics

Metric	Value
Authenticated Packets	40
Blocked Packets	1948
Replay Attacks Blocked	1
Authentication Success Rate	~2%

The following table summarizes the cryptographic performance of the ChaCha20-Poly1305 algorithm in terms of encryption and decryption time, as well as processing overhead.

Table 8: Cryptographic Performance of ChaCha20-Poly1305

Operation	Time
Encryption	91.6 μ s
Decryption	16.24 μ s
Security Overhead per Packet	28 bytes
Processing Impact	Low

The following table illustrates the additional overhead introduced by the security mechanism, including per-packet and overall network overhead.

Table 9: Security Overhead Analysis

Metric	Value	Description
Security Overhead per Packet	28 bytes	Added due to nonce + tag
Network Overhead	1400 bytes	Total additional transmission cost

The following table shows the impact of the security implementation on system energy consumption and overall efficiency.

Table 10: Energy Efficiency Evaluation

Metric	Value	Description
Final Energy	99.551 J	Remaining system energy
Efficiency	99.551%	Very low energy consumption impact

The following table provides a summary of attack detection results under different scenarios, including normal traffic, jamming, and replay attacks.

Table 11: Attack Detection Summary under Different Scenarios

Attack Type	Result	Count
Normal Traffic	Accepted	40
Jamming/Corrupted Packets	Blocked	1948
Replay Attack	Blocked	1

4.11 Chapter Summary

This chapter presented the practical implementation of the proposed UAV communication system. The ns-3 simulation environment successfully demonstrated secure and efficient communication between GCS and UAVs using a MAVLink-like protocol enhanced with ChaCha20-Poly1305 encryption.

Conclusion

This work presented the design and implementation of a secure UAV communication system based on a MAVLink-like protocol enhanced with ChaCha20-Poly1305 authenticated encryption. The primary objective was to address the security limitations of conventional MAVLink communication, particularly the lack of confidentiality, integrity, and authentication in UAV wireless networks.

A lightweight security layer was implemented at the application level, combining ChaCha20 encryption with Poly1305 authentication and sequence number tracking to mitigate replay attacks. The system was modeled using the OSI reference architecture and deployed over an IPv4-based IEEE 802.11b ad-hoc network using UDP for UAV–GCS communication.

The proposed system was evaluated using the ns-3 simulation environment under multiple scenarios, including normal operation and malicious attack conditions. The results demonstrated that the system effectively detects and blocks unauthorized and replayed packets while maintaining reliable communication performance.

Performance evaluation showed minimal overhead in terms of latency, throughput, and energy consumption, confirming the suitability of ChaCha20-Poly1305 for resource-constrained UAV environments. The cryptographic operations also exhibited low processing cost, ensuring real-time applicability.

Despite the promising results, the study is limited to simulation-based validation and simplified key management assumptions. Future improvements are required to extend the system beyond simulation environments.

In conclusion, the proposed system provides an efficient and lightweight security framework for MAVLink-based UAV communications, achieving a balanced trade-off between security strength and system performance for real-time aerial applications.

Future Work

Although the proposed system demonstrates strong performance and security in simulation, several enhancements can be considered for future development. First, integrating a dynamic key exchange mechanism would improve overall security by eliminating reliance on pre-shared keys. Second, extending the system to support large-scale UAV swarms would allow evaluation of scalability under dense network conditions.

Additionally, implementing and testing the proposed security framework on real UAV hardware would provide more realistic validation under physical channel conditions, including interference, mobility, and packet loss. Further optimization of energy consumption can also be explored to extend UAV operational lifetime during long missions.

Finally, future research may focus on enhancing attack detection capabilities by incorporating advanced intrusion detection mechanisms for UAV networks to improve resilience against sophisticated cyber threats.

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